Assignment 7: Sample Data Insertion

Hospital Management System - realistic sample data for all tables

1. Objective

This assignment populates all ten tables with realistic sample data for a mid-sized city hospital (Bengaluru), covering September to November 2026. The data volume is deliberately small enough to inspect, but rich enough to exercise every relationship: 15 patients, 10 doctors in 6 departments, 20 appointments (completed, cancelled and upcoming), 8 admissions (one patient still admitted), 9 bills (inpatient and outpatient, paid, partial and pending), 12 medical records, 10 prescriptions and 18 prescription line items.

2. Insertion Order

Parent tables are loaded before child tables so foreign keys are always satisfied: DEPARTMENTS, PATIENTS, DOCTOR, APPOINTMENT, ADMISSIONS, MEDICAL_RECORD, BILL, PRESCRIPTION, MEDICINES, PRESCRIPTION_MEDICINE. Note that MEDICINES must exist before PRESCRIPTION_MEDICINE. All values obey the constraints from Assignment 6 (valid statuses, discharge after admission, positive prices and quantities). Patient names, phone numbers and e-mail addresses are fictional.

3. INSERT Statements

DEPARTMENTS (6 rows)

```
INSERT INTO DEPARTMENTS
    (department_id, department_name, location)
VALUES
    (1, 'Cardiology', 'Block A, Floor 2'),
    (2, 'Neurology', 'Block B, Floor 3'),
    (3, 'Orthopedics', 'Block A, Floor 1'),
    (4, 'Pediatrics', 'Block C, Floor 1'),
    (5, 'General Medicine', 'Block B, Floor 1'),
    (6, 'Gynecology', 'Block C, Floor 2');
```

PATIENTS (15 rows)

```
INSERT INTO PATIENTS
    (patient_id, name, DOB, Gender, ph_no, address)
VALUES
    (1001, 'Ravi Shankar', '1985-03-12', 'Male', '9845010001', 'Jayanagar, Bengaluru'),
    (1002, 'Anitha Rao', '1992-07-25', 'Female', '9845010002', 'Anekal, Bengaluru'),
    (1003, 'Mohan Das', '1958-11-02', 'Male', '9845010003', 'Hebbagodi, Bengaluru'),
    (1004, 'Sneha Kulkarni', '2001-01-18', 'Female', '9845010004', 'Electronic City, Bengaluru'),
    (1005, 'Imran Pasha', '1979-09-30', 'Male', '9845010005', 'Attibele, Bengaluru'),
    (1006, 'Divya Hegde', '1995-05-14', 'Female', '9845010006', 'Koramangala, Bengaluru'),
    (1007, 'Karthik Naidu', '2015-08-09', 'Male', '9845010007', 'Jigani, Bengaluru'),
    (1008, 'Fatima Begum', '1968-12-21', 'Female', '9845010008', 'Chandapura, Bengaluru'),
    (1009, 'Suresh Babu', '1949-04-05', 'Male', '9845010009', 'Bommasandra, Bengaluru'),
    (1010, 'Pooja Sharma', '1988-02-27', 'Female', '9845010010', 'HSR Layout, Bengaluru'),
    (1011, 'Naveen Gowda', '1999-10-10', 'Male', '9845010011', 'Sarjapur, Bengaluru'),
    (1012, 'Lakshmi Devi', '1975-06-17', 'Female', '9845010012', 'Anekal, Bengaluru'),
    (1013, 'Aarav Menon', '2019-03-03', 'Male', '9845010013', 'Whitefield, Bengaluru'),
    (1014, 'Rekha Joshi', '1983-09-09', 'Female', '9845010014', 'Indiranagar, Bengaluru'),
    (1015, 'Vikram Singh', '1971-01-29', 'Male', '9845010015', 'BTM Layout, Bengaluru');
```

DOCTOR (10 rows)

```
INSERT INTO DOCTOR
    (doctor_id, doctor_name, specialization, ph_no, email, department_id)
VALUES
    (101, 'Dr. Anil Kumar', 'Cardiologist', '8041020001', 'anil.kumar@cityhospital.in', 1),
    (102, 'Dr. Meera Nair', 'Cardiologist', '8041020002', 'meera.nair@cityhospital.in', 1),
    (103, 'Dr. Suresh Rao', 'Neurologist', '8041020003', 'suresh.rao@cityhospital.in', 2),
    (104, 'Dr. Kavita Shetty', 'Neurologist', '8041020004', 'kavita.shetty@cityhospital.in', 2),
    (105, 'Dr. Rajesh Gowda', 'Orthopedic Surgeon', '8041020005', 'rajesh.gowda@cityhospital.in', 3),
    (106, 'Dr. Priya Menon', 'Pediatrician', '8041020006', 'priya.menon@cityhospital.in', 4),
    (107, 'Dr. Arjun Reddy', 'Pediatrician', '8041020007', 'arjun.reddy@cityhospital.in', 4),
    (108, 'Dr. Lakshmi Iyer', 'General Physician', '8041020008', 'lakshmi.iyer@cityhospital.in', 5),
```

```
    (109, 'Dr. Mohammed Faiz', 'General Physician', '8041020009', 'mohammed.faiz@cityhospital.in', 5),
    (110, 'Dr. Sunita Patil', 'Gynecologist', '8041020010', 'sunita.patil@cityhospital.in', 6);
```

APPOINTMENT (20 rows)

```
INSERT INTO APPOINTMENT
    (appointment_id, appointment_date, appointment_status, patient_id, doctor_id)
VALUES
    (5001, '2026-09-02', 'Completed', 1001, 101),
    (5002, '2026-09-03', 'Completed', 1003, 101),
    (5003, '2026-09-05', 'Completed', 1004, 108),
    (5004, '2026-09-08', 'Completed', 1007, 106),
    (5005, '2026-09-10', 'Cancelled', 1005, 105),
    (5006, '2026-09-12', 'Completed', 1009, 102),
    (5007, '2026-09-15', 'Completed', 1010, 110),
    (5008, '2026-09-18', 'Completed', 1002, 108),
    (5009, '2026-09-20', 'Completed', 1013, 107),
    (5010, '2026-09-22', 'Completed', 1008, 103),
    (5011, '2026-09-25', 'Completed', 1011, 109),
    (5012, '2026-09-28', 'Cancelled', 1006, 108),
    (5013, '2026-10-01', 'Completed', 1001, 101),
    (5014, '2026-10-03', 'Completed', 1012, 104),
    (5015, '2026-10-06', 'Completed', 1014, 110),
    (5016, '2026-10-09', 'Completed', 1015, 105),
    (5017, '2026-10-26', 'Scheduled', 1003, 102),
    (5018, '2026-10-28', 'Scheduled', 1007, 106),
    (5019, '2026-10-30', 'Scheduled', 1009, 101),
    (5020, '2026-11-02', 'Scheduled', 1004, 108);
```

ADMISSIONS (8 rows)

```
INSERT INTO ADMISSIONS
    (admission_id, admission_date, discharge_date, room_number, patient_id, doctor_id)
VALUES
    (7001, '2026-09-03', '2026-09-09', 'A-201', 1003, 101),
    (7002, '2026-09-12', '2026-09-16', 'A-205', 1009, 102),
    (7003, '2026-09-14', '2026-09-20', 'B-310', 1008, 103),
    (7004, '2026-09-18', '2026-09-21', 'C-105', 1007, 106),
    (7005, '2026-10-01', '2026-10-07', 'A-102', 1005, 105),
    (7006, '2026-10-08', '2026-10-12', 'C-210', 1010, 110),
    (7007, '2026-10-16', '2026-10-19', 'B-112', 1011, 109),
    (7008, '2026-10-20', NULL, 'B-315', 1012, 104);
```

MEDICAL_RECORD (12 rows)

```
INSERT INTO MEDICAL_RECORD
    (record_id, diagnosis, treatment, record_date, patient_id, doctor_id)
VALUES
    (3001, 'Hypertension', 'Antihypertensive therapy and diet control', '2026-09-02', 1001, 101),
    (3002, 'Myocardial infarction', 'Angioplasty and cardiac monitoring', '2026-09-03', 1003, 101),
    (3003, 'Viral fever', 'Rest, hydration and antipyretics', '2026-09-05', 1004, 108),
    (3004, 'Pediatric pneumonia', 'Antibiotics and oxygen support', '2026-09-18', 1007, 106),
    (3005, 'Coronary artery disease', 'Medication and lifestyle changes', '2026-09-12', 1009, 102),
    (3006, 'Ischemic stroke', 'Thrombolysis and physiotherapy', '2026-09-14', 1008, 103),
    (3007, 'Femur fracture', 'Surgical fixation', '2026-10-01', 1005, 105),
    (3008, 'Gestational diabetes', 'Diet plan and insulin', '2026-10-08', 1010, 110),
    (3009, 'Typhoid', 'Antibiotic course', '2026-10-16', 1011, 109),
    (3010, 'Migraine', 'Preventive therapy', '2026-10-03', 1012, 104),
    (3011, 'Hypertension', 'Medication adjustment', '2026-10-01', 1001, 101),
    (3012, 'Type 2 diabetes', 'Metformin and diet plan', '2026-09-18', 1002, 108);
```

BILL (9 rows)

```
INSERT INTO BILL
    (bill_id, total_amount, payment_status, bill_date, patient_id, admission_id)
VALUES
    (9001, 85000.00, 'Paid', '2026-09-09', 1003, 7001),
    (9002, 62500.50, 'Paid', '2026-09-16', 1009, 7002),
    (9003, 48000.00, 'Partial', '2026-09-20', 1008, 7003),
    (9004, 15750.00, 'Paid', '2026-09-21', 1007, 7004),
    (9005, 120000.00, 'Pending', '2026-10-07', 1005, 7005),
    (9006, 54300.00, 'Paid', '2026-10-12', 1010, 7006),
    (9007, 18900.00, 'Pending', '2026-10-19', 1011, 7007),
    (9008, 1500.00, 'Paid', '2026-09-05', 1004, NULL),
```

```
    (9009, 800.00, 'Paid', '2026-09-18', 1002, NULL);
```

PRESCRIPTION (10 rows)

```
INSERT INTO PRESCRIPTION
    (prescription_id, prescription_date, patient_id, doctor_id)
VALUES
    (4001, '2026-09-02', 1001, 101),
    (4002, '2026-09-03', 1003, 101),
    (4003, '2026-09-05', 1004, 108),
    (4004, '2026-09-18', 1007, 106),
    (4005, '2026-09-12', 1009, 102),
    (4006, '2026-10-08', 1010, 110),
    (4007, '2026-10-16', 1011, 109),
    (4008, '2026-10-03', 1012, 104),
    (4009, '2026-10-01', 1001, 101),
    (4010, '2026-09-18', 1002, 108);
```

MEDICINES (11 rows)

```
INSERT INTO MEDICINES
    (medicine_id, medicine_name, manufacturer, price, stock_quantity)
VALUES
    (201, 'Paracetamol 500mg', 'Cipla', 2.50, 5000),
    (202, 'Amoxicillin 500mg', 'Sun Pharma', 8.00, 2500),
    (203, 'Atorvastatin 20mg', 'Dr. Reddys', 12.00, 1800),
    (204, 'Amlodipine 5mg', 'Lupin', 4.50, 3000),
    (205, 'Metformin 500mg', 'Zydus', 3.00, 4000),
    (206, 'Aspirin 75mg', 'Bayer', 1.80, 6000),
    (207, 'Clopidogrel 75mg', 'Torrent', 9.50, 150),
    (208, 'Ceftriaxone 1g Inj', 'Alkem', 45.00, 400),
    (209, 'Insulin Glargine', 'Biocon', 650.00, 80),
    (210, 'Ibuprofen 400mg', 'Abbott', 3.20, 3500),
    (211, 'Omeprazole 20mg', 'Dr. Reddys', 5.00, 2000);
```

PRESCRIPTION_MEDICINE (18 rows)

```
INSERT INTO PRESCRIPTION_MEDICINE
    (prescription_id, medicine_id, quantity)
VALUES
    (4001, 204, 30),
    (4001, 206, 30),
    (4002, 206, 30),
    (4002, 207, 30),
    (4002, 203, 30),
    (4003, 201, 10),
    (4004, 202, 14),
    (4004, 201, 10),
    (4005, 203, 30),
    (4005, 206, 30),
    (4005, 204, 30),
    (4006, 209, 2),
    (4006, 205, 60),
    (4007, 208, 6),
    (4007, 201, 10),
    (4008, 210, 10),
    (4009, 204, 30),
    (4010, 205, 60);
```

4. Verification of Inserted Data

4.1 Row count per table

```
SELECT 'DEPARTMENTS' AS table_name, COUNT(*) AS row_count FROM DEPARTMENTS UNION ALL
SELECT 'PATIENTS' AS table_name, COUNT(*) AS row_count FROM PATIENTS UNION ALL
SELECT 'DOCTOR' AS table_name, COUNT(*) AS row_count FROM DOCTOR UNION ALL
SELECT 'APPOINTMENT' AS table_name, COUNT(*) AS row_count FROM APPOINTMENT UNION ALL
SELECT 'ADMISSIONS' AS table_name, COUNT(*) AS row_count FROM ADMISSIONS UNION ALL
SELECT 'MEDICAL_RECORD' AS table_name, COUNT(*) AS row_count FROM MEDICAL_RECORD UNION ALL
SELECT 'BILL' AS table_name, COUNT(*) AS row_count FROM BILL UNION ALL
SELECT 'PRESCRIPTION' AS table_name, COUNT(*) AS row_count FROM PRESCRIPTION UNION ALL
SELECT 'MEDICINES' AS table_name, COUNT(*) AS row_count FROM MEDICINES UNION ALL
SELECT 'PRESCRIPTION_MEDICINE' AS table_name, COUNT(*) AS row_count FROM PRESCRIPTION_MEDICINE;
```

table_name	row_count
DEPARTMENTS	6
PATIENTS	15
DOCTOR	10
APPOINTMENT	20
ADMISSIONS	8
MEDICAL_RECORD	12
BILL	9
PRESCRIPTION	10
MEDICINES	11
PRESCRIPTION_MEDICINE	18

10 row(s) returned

4.2 Table contents

```
SELECT * FROM DEPARTMENTS;
```

department_id	department_name	location
1	Cardiology	Block A, Floor 2
2	Neurology	Block B, Floor 3
3	Orthopedics	Block A, Floor 1
4	Pediatrics	Block C, Floor 1
5	General Medicine	Block B, Floor 1
6	Gynecology	Block C, Floor 2

6 row(s) returned

```
SELECT * FROM PATIENTS;
```

patient_id	name	DOB	Gender	ph_no	address
1001	Ravi Shankar	1985-03-12	Male	9845010001	Jayanagar, Bengaluru
1002	Anitha Rao	1992-07-25	Female	9845010002	Anekal, Bengaluru
1003	Mohan Das	1958-11-02	Male	9845010003	Hebbagodi, Bengaluru
1004	Sneha Kulkarni	2001-01-18	Female	9845010004	Electronic City, Bengaluru
1005	Imran Pasha	1979-09-30	Male	9845010005	Attibele, Bengaluru
1006	Divya Hegde	1995-05-14	Female	9845010006	Koramangala, Bengaluru
1007	Karthik Naidu	2015-08-09	Male	9845010007	Jigani, Bengaluru
1008	Fatima Begum	1968-12-21	Female	9845010008	Chandapura, Bengaluru
1009	Suresh Babu	1949-04-05	Male	9845010009	Bommasandra, Bengaluru
1010	Pooja Sharma	1988-02-27	Female	9845010010	HSR Layout, Bengaluru
1011	Naveen Gowda	1999-10-10	Male	9845010011	Sarjapur, Bengaluru
1012	Lakshmi Devi	1975-06-17	Female	9845010012	Anekal, Bengaluru
1013	Aarav Menon	2019-03-03	Male	9845010013	Whitefield, Bengaluru
1014	Rekha Joshi	1983-09-09	Female	9845010014	Indiranagar, Bengaluru
1015	Vikram Singh	1971-01-29	Male	9845010015	BTM Layout, Bengaluru

15 row(s) returned

```
SELECT * FROM DOCTOR;
```

doctor_id	doctor_name	specialization	ph_no	email	department_id
101	Dr. Anil Kumar	Cardiologist	8041020001	anil.kumar@cityhospital.in	1
102	Dr. Meera Nair	Cardiologist	8041020002	meera.nair@cityhospital.in	1
103	Dr. Suresh Rao	Neurologist	8041020003	suresh.rao@cityhospital.in	2
104	Dr. Kavita Shetty	Neurologist	8041020004	kavita.shetty@cityhospital.in	2
105	Dr. Rajesh Gowda	Orthopedic Surgeon	8041020005	rajesh.gowda@cityhospital.in	3
106	Dr. Priya Menon	Pediatrician	8041020006	priya.menon@cityhospital.in	4
107	Dr. Arjun Reddy	Pediatrician	8041020007	arjun.reddy@cityhospital.in	4
108	Dr. Lakshmi Iyer	General Physician	8041020008	lakshmi.iyer@cityhospital.in	5
109	Dr. Mohammed Faiz	General Physician	8041020009	mohammed.faiz@cityhospital.in	5
110	Dr. Sunita Patil	Gynecologist	8041020010	sunita.patil@cityhospital.in	6

10 row(s) returned

```
SELECT * FROM APPOINTMENT;
```

appointment_id	appointment_date	appointment_status	patient_id	doctor_id

5001	2026-09-02	Completed	1001	101
5002	2026-09-03	Completed	1003	101
5003	2026-09-05	Completed	1004	108
5004	2026-09-08	Completed	1007	106
5005	2026-09-10	Cancelled	1005	105
5006	2026-09-12	Completed	1009	102
5007	2026-09-15	Completed	1010	110
5008	2026-09-18	Completed	1002	108
5009	2026-09-20	Completed	1013	107
5010	2026-09-22	Completed	1008	103
5011	2026-09-25	Completed	1011	109
5012	2026-09-28	Cancelled	1006	108
5013	2026-10-01	Completed	1001	101
5014	2026-10-03	Completed	1012	104
5015	2026-10-06	Completed	1014	110
5016	2026-10-09	Completed	1015	105
5017	2026-10-26	Scheduled	1003	102
5018	2026-10-28	Scheduled	1007	106
5019	2026-10-30	Scheduled	1009	101
5020	2026-11-02	Scheduled	1004	108

20 row(s) returned

```
SELECT * FROM ADMISSIONS;
```

admission_id	admission_date	discharge_date	room_number	patient_id	doctor_id
7001	2026-09-03	2026-09-09	A-201	1003	101
7002	2026-09-12	2026-09-16	A-205	1009	102
7003	2026-09-14	2026-09-20	B-310	1008	103
7004	2026-09-18	2026-09-21	C-105	1007	106
7005	2026-10-01	2026-10-07	A-102	1005	105
7006	2026-10-08	2026-10-12	C-210	1010	110
7007	2026-10-16	2026-10-19	B-112	1011	109
7008	2026-10-20	NULL	B-315	1012	104

8 row(s) returned

```
SELECT * FROM MEDICAL_RECORD;
```

record_id	diagnosis	treatment	record_date	patient_id	doctor_id
3001	Hypertension	Antihypertensive therapy and diet control	2026-09-02	1001	101
3002	Myocardial infarction	Angioplasty and cardiac monitoring	2026-09-03	1003	101
3003	Viral fever	Rest, hydration and antipyretics	2026-09-05	1004	108
3004	Pediatric pneumonia	Antibiotics and oxygen support	2026-09-18	1007	106
3005	Coronary artery disease	Medication and lifestyle changes	2026-09-12	1009	102
3006	Ischemic stroke	Thrombolysis and physiotherapy	2026-09-14	1008	103
3007	Femur fracture	Surgical fixation	2026-10-01	1005	105
3008	Gestational diabetes	Diet plan and insulin	2026-10-08	1010	110
3009	Typhoid	Antibiotic course	2026-10-16	1011	109
3010	Migraine	Preventive therapy	2026-10-03	1012	104
3011	Hypertension	Medication adjustment	2026-10-01	1001	101
3012	Type 2 diabetes	Metformin and diet plan	2026-09-18	1002	108

12 row(s) returned

```
SELECT * FROM BILL;
```

bill_id	total_amount	payment_status	bill_date	patient_id	admission_id
9001	85000.00	Paid	2026-09-09	1003	7001
9002	62500.50	Paid	2026-09-16	1009	7002
9003	48000.00	Partial	2026-09-20	1008	7003
9004	15750.00	Paid	2026-09-21	1007	7004
9005	120000.00	Pending	2026-10-07	1005	7005
9006	54300.00	Paid	2026-10-12	1010	7006
9007	18900.00	Pending	2026-10-19	1011	7007
9008	1500.00	Paid	2026-09-05	1004	NULL

9009	800.00	Paid	2026-09-18	1002	NULL

9 row(s) returned

```
SELECT * FROM PRESCRIPTION;
```

prescription_id	prescription_date	patient_id	doctor_id
4001	2026-09-02	1001	101
4002	2026-09-03	1003	101
4003	2026-09-05	1004	108
4004	2026-09-18	1007	106
4005	2026-09-12	1009	102
4006	2026-10-08	1010	110
4007	2026-10-16	1011	109
4008	2026-10-03	1012	104
4009	2026-10-01	1001	101
4010	2026-09-18	1002	108

10 row(s) returned

```
SELECT * FROM MEDICINES;
```

medicine_id	medicine_name	manufacturer	price	stock_quantity
201	Paracetamol 500mg	Cipla	2.50	5000
202	Amoxicillin 500mg	Sun Pharma	8.00	2500
203	Atorvastatin 20mg	Dr. Reddys	12.00	1800
204	Amlodipine 5mg	Lupin	4.50	3000
205	Metformin 500mg	Zydus	3.00	4000
206	Aspirin 75mg	Bayer	1.80	6000
207	Clopidogrel 75mg	Torrent	9.50	150
208	Ceftriaxone 1g Inj	Alkem	45.00	400
209	Insulin Glargine	Biocon	650.00	80
210	Ibuprofen 400mg	Abbott	3.20	3500
211	Omeprazole 20mg	Dr. Reddys	5.00	2000

11 row(s) returned

```
SELECT * FROM PRESCRIPTION_MEDICINE;
```

prescription_id	medicine_id	quantity
4001	204	30
4001	206	30
4002	206	30
4002	207	30
4002	203	30
4003	201	10
4004	202	14
4004	201	10
4005	203	30
4005	206	30
4005	204	30
4006	209	2
4006	205	60
4007	208	6
4007	201	10
4008	210	10
4009	204	30
4010	205	60

18 row(s) returned

4.3 Integrity confirmation

All inserts succeeded without a constraint error. The engine's own foreign-key check confirms no orphan references:

```
PRAGMA foreign_key_check;
```

table	rowid	parent	fkid

(no rows)		

0 row(s) returned

Data features worth noting: BILL rows 9008 and 9009 are outpatient bills (admission_id is NULL); admission 7008 has no discharge_date because the patient is still admitted; appointments 5005 and 5012 are Cancelled and 5017-5020 are Scheduled for the future; medicine 211 (Omeprazole) is in stock but has never been prescribed, and medicines 207 and 209 have low stock - these variations are used in the reports of Assignments 8 and 9.